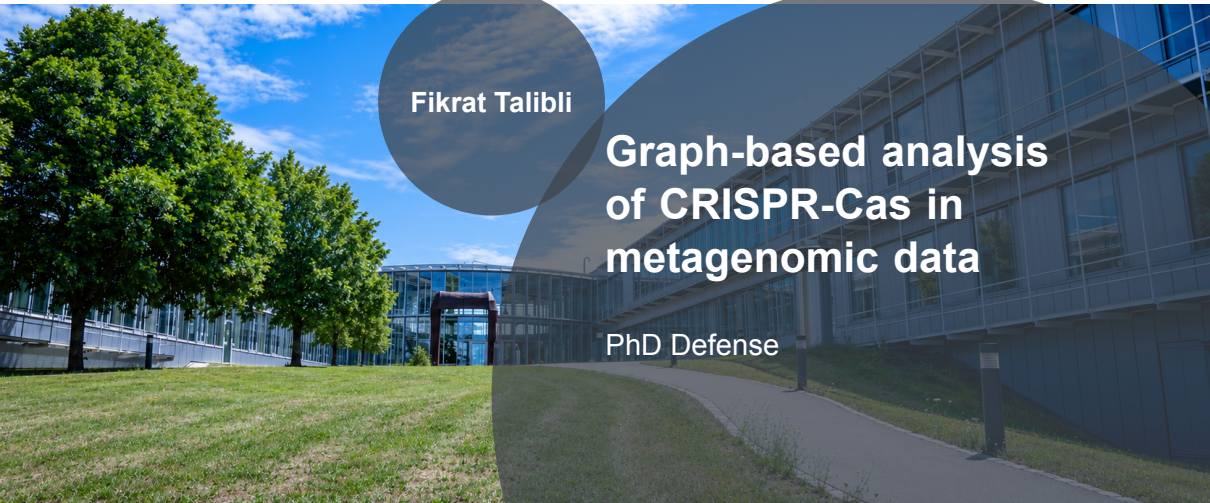
University of Stuttgart
Institute of Biomedical Genetics



Fikrat Talibli

Graph-based analysis of CRISPR-Cas in metagenomic data

PhD Defense

Contents

- Motivation and overview
- Results I: CRISPR array detection
- Results II: CAS cassette classification
- Results III: Protospacer extension
- Conclusions and outlook

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CRISPR-Cas immune response

- Bacteria and viruses/bacteriophages are constantly in an evolutionary arms race
- Bacteria have evolved immune systems such as **CRISPR-Cas**
- **C**lustered **R**egularly **I**nterspaced **S**hort **P**alindromic **R**epeats
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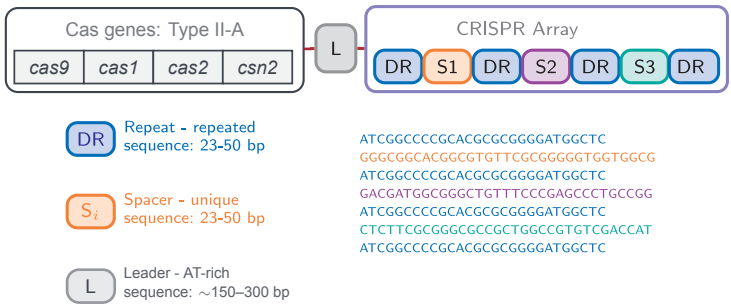

Motivation

Diversity of CRISPR-Cas systems

- Many known systems of different classes, types and subtypes
- Limited to cultivable bacteria and archaea — less than 5%

Metagenomics

- Analysis of the DNA of a whole microbiome
- Millions of 100–150 nt reads

Soil sample

DNA sequencing

Reads

R1 ACTG**TCTGAAG** ...
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R3 GGCATACCGTT ...
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Assembly of reads into longer stretches (contigs): ~25% data loss

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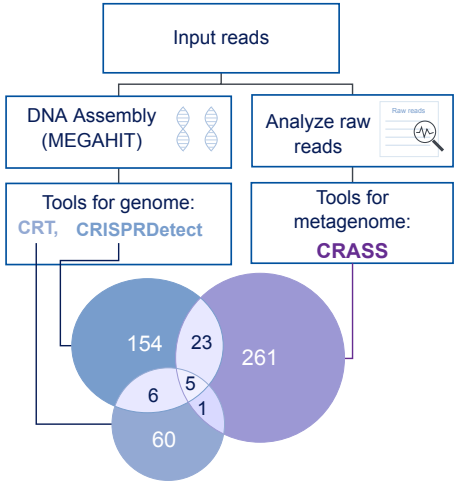

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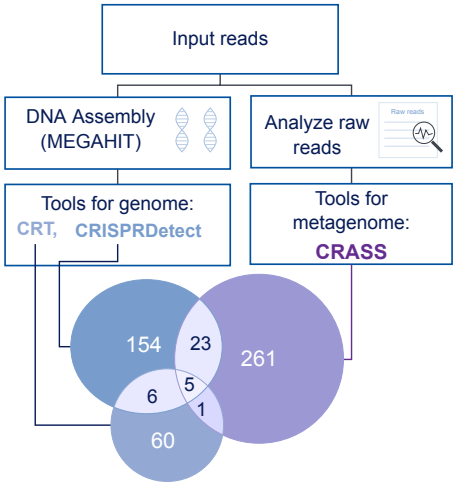

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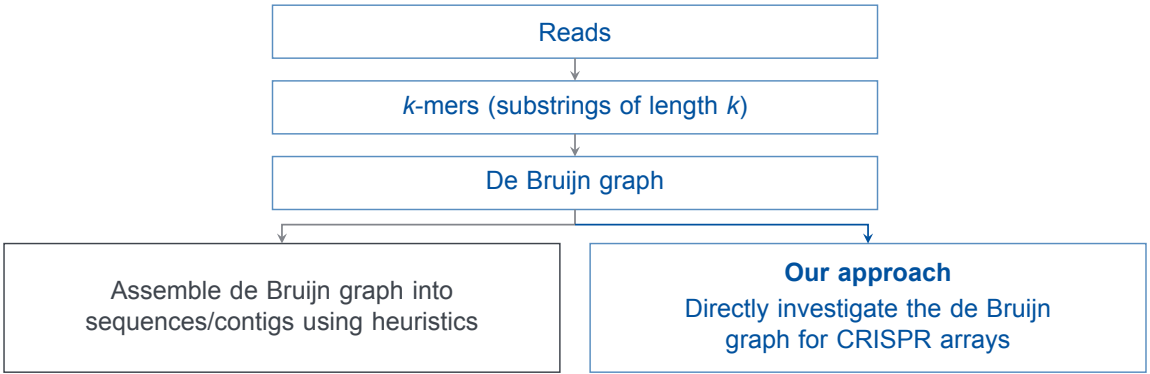
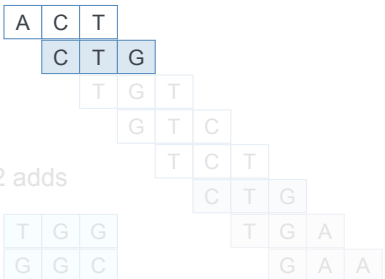


De Bruijn graph: a toy example

Reads (10 bases each)

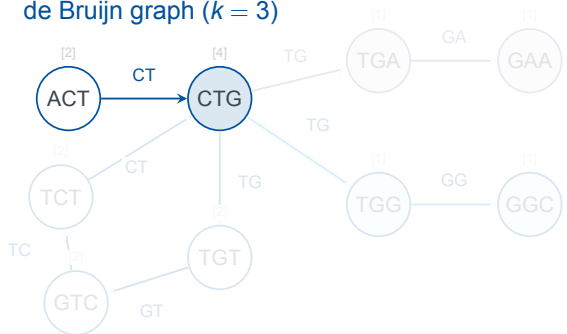
R1	A	C	T	G	T	C	T	G	A	A
R2	A	C	T	G	T	C	T	G	G	C

Overlapping 3-mers from read 1



Read 2 adds

de Bruijn graph ($k = 3$)



ACT → CTG: 2-base overlap CT

Node counts in brackets: 3-mer occurrences across both reads.

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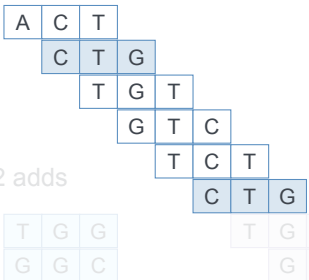

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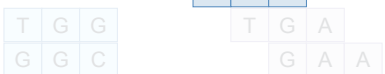

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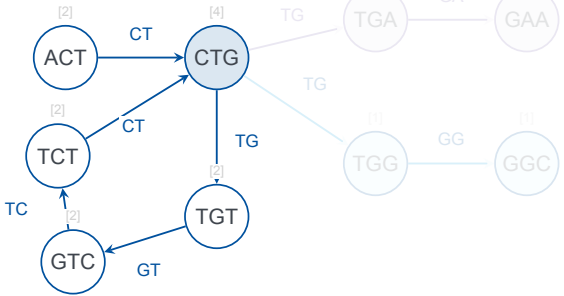

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	A	C	T							
		C	T	G						
			T	G	T					
				G	T	C				
					T	C	T			
						C	T	G		
							T	G	A	
								G	A	A

Read 2 adds

	T	G	G							
	G	G	C							

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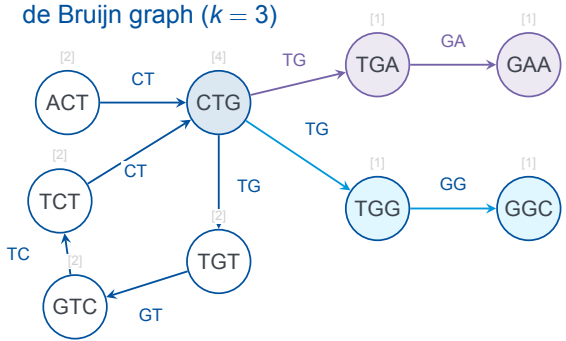

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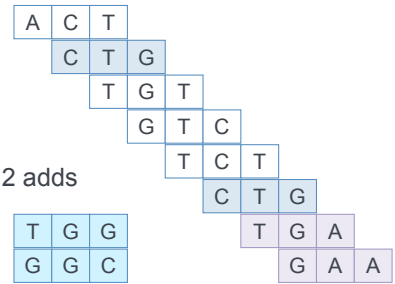

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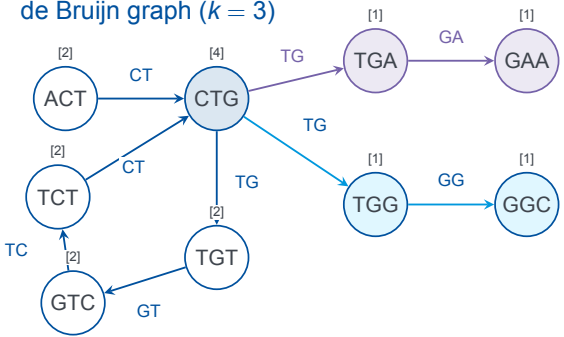

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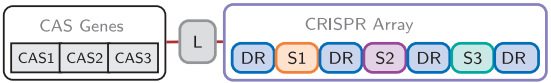
CRISPR arrays in de Bruijn graphs



DR Repeat – repeated sequence: 23-50 bp

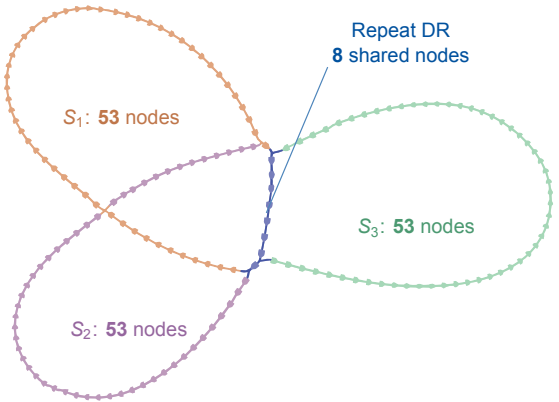
S_i Spacer – unique sequence: 23-50 bp

```
ATCGGCCCCGCACGCGCGGGGATGGCTC
GGGCGGCACGGCGTGTTCGCGGGGGTGGTGGCG
ATCGGCCCCGCACGCGCGGGGATGGCTC
GACGATGGCGGGCTGTTTCCCGAGCCCTGCCGG
ATCGGCCCCGCACGCGCGGGGATGGCTC
CTCTTCGCGGGCGCCGCTGGCCGTGTCGACCAT
ATCGGCCCCGCACGCGCGGGGATGGCTC
```

Sequence	Length
Repeat DR	28 bp
Spacer S_i	33 bp

- Repeat nodes are shared
- Spacer nodes are unique

de Bruijn graph ($k = 21$)



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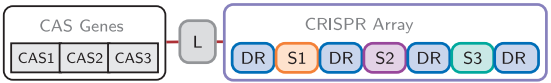

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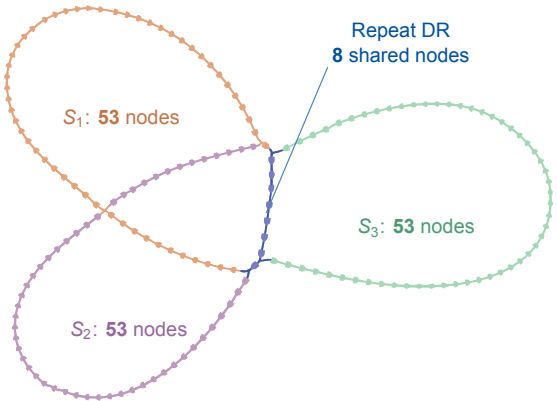
de Bruijn graph ($k = 21$)



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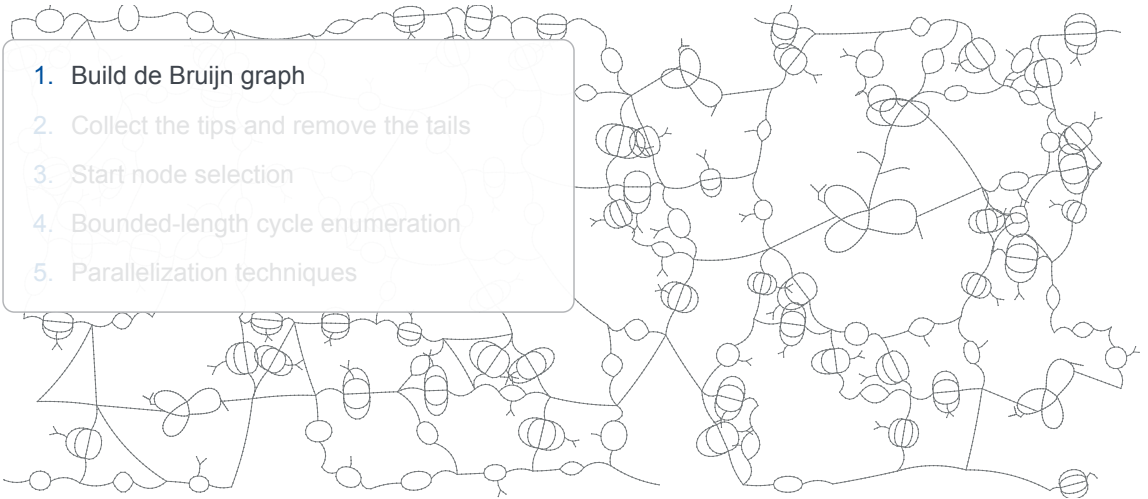

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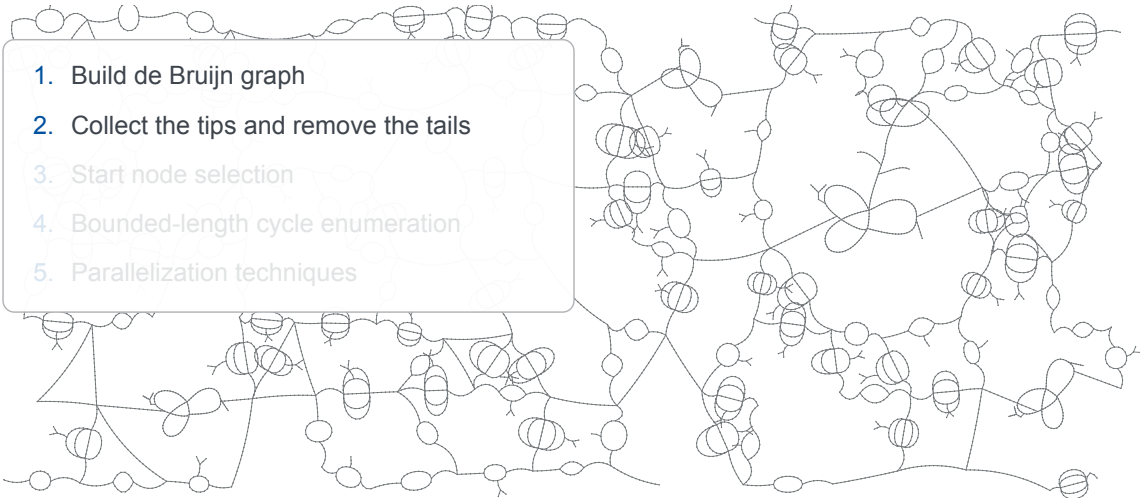

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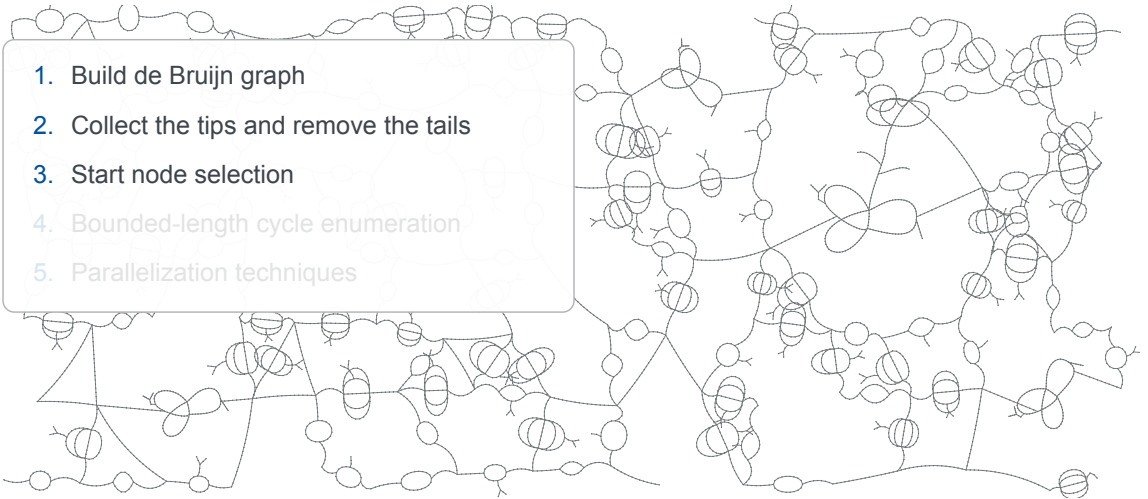

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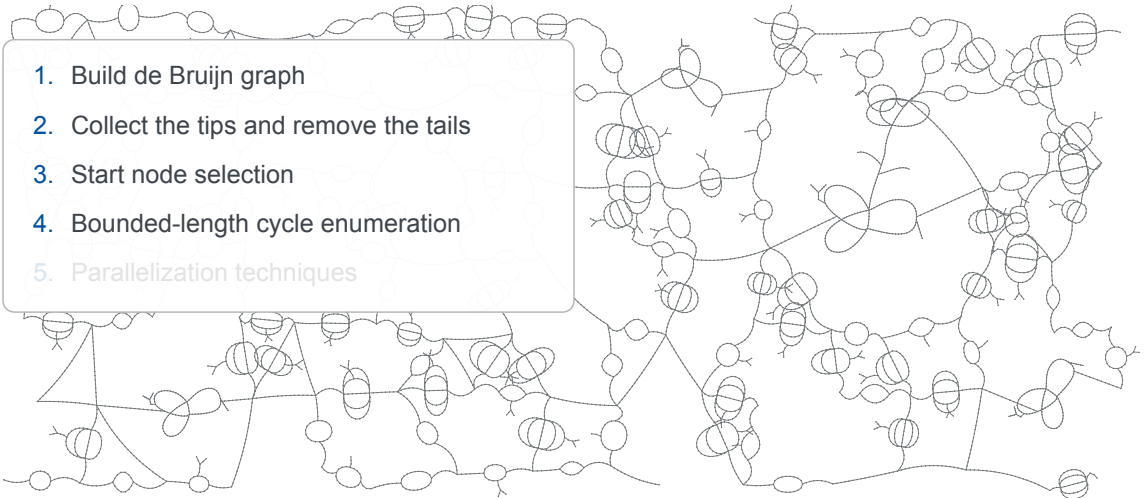

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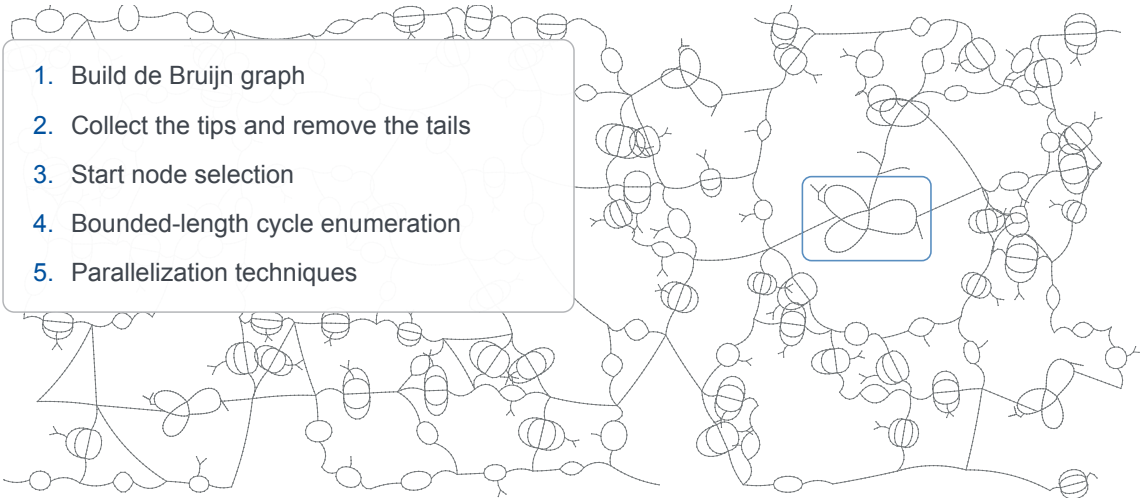

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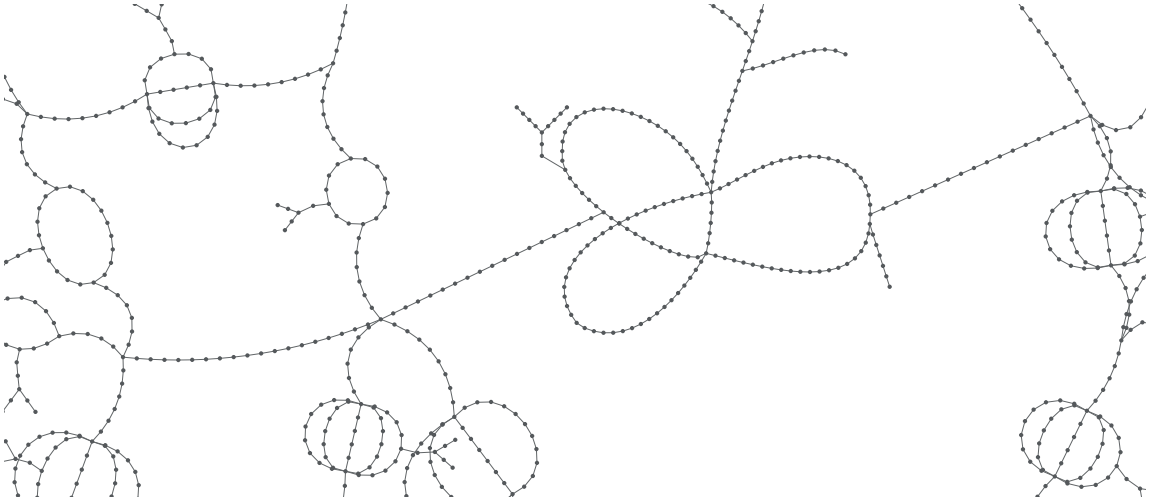

CRISPR array detection

Tip removal

tips : $\deg^+(v) = 0$

for each tip v:
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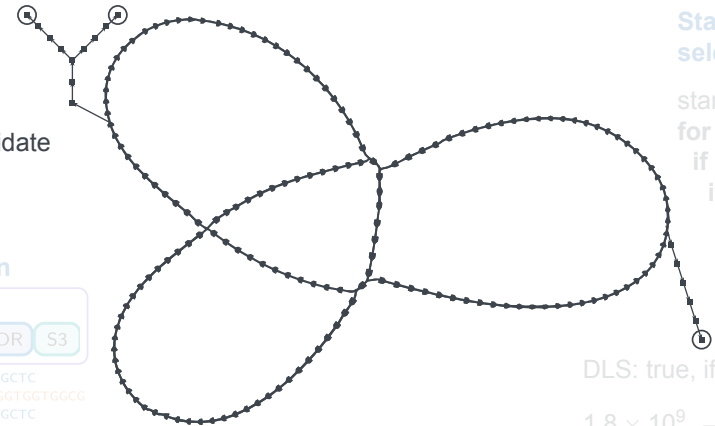

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Cycle enumeration

CRISPR Array

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DLS: true, if start = end

$1.8 \times 10^9 \longrightarrow 2.4 \times 10^4$

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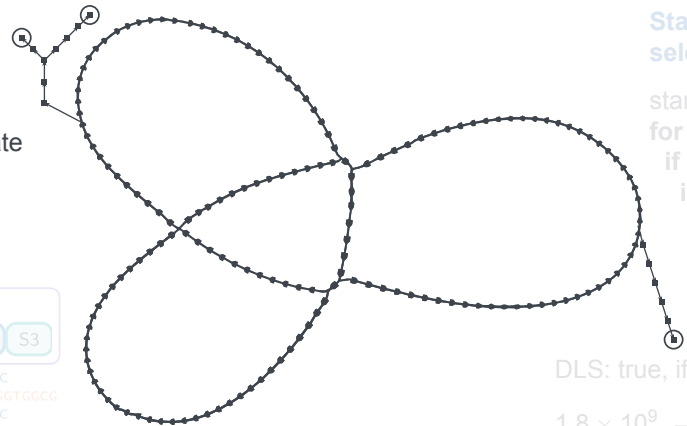

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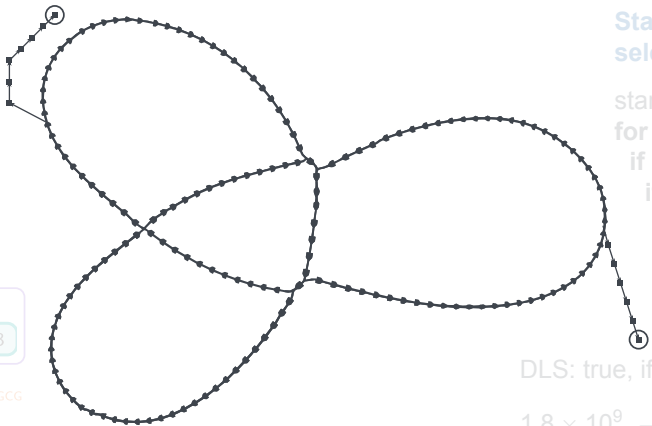

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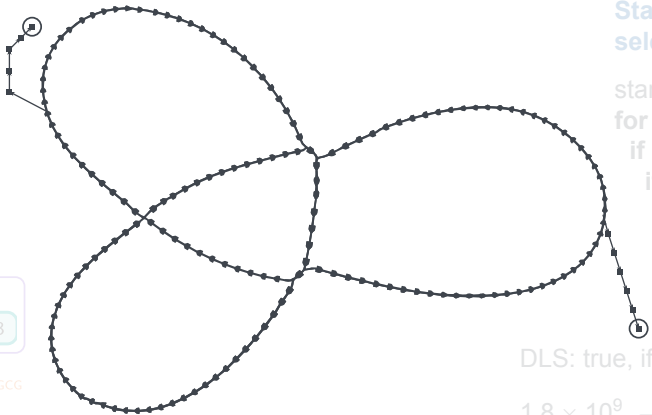

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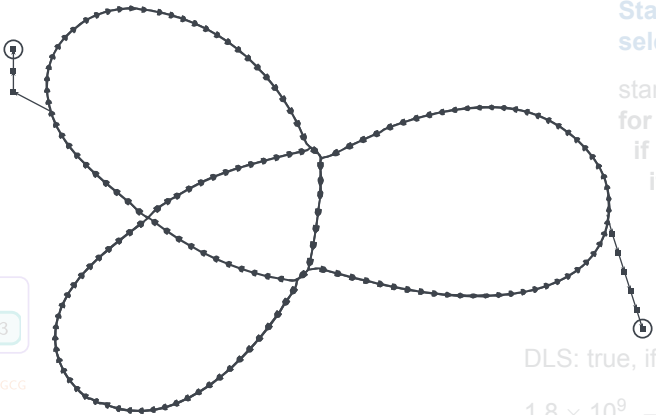

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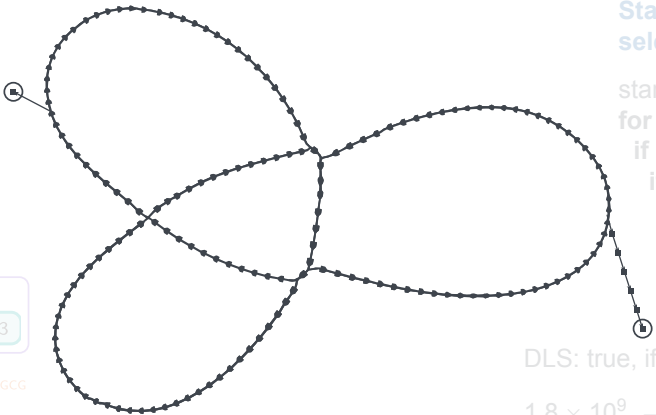

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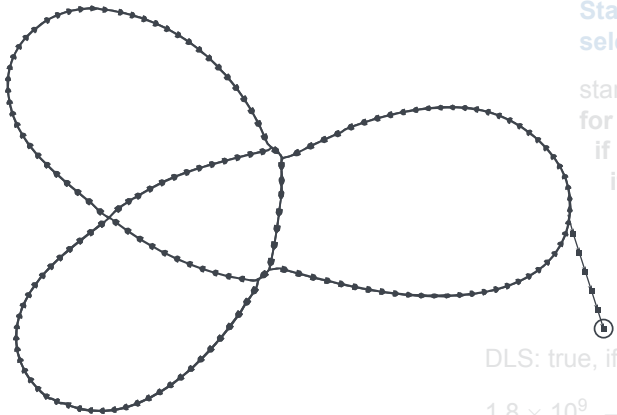

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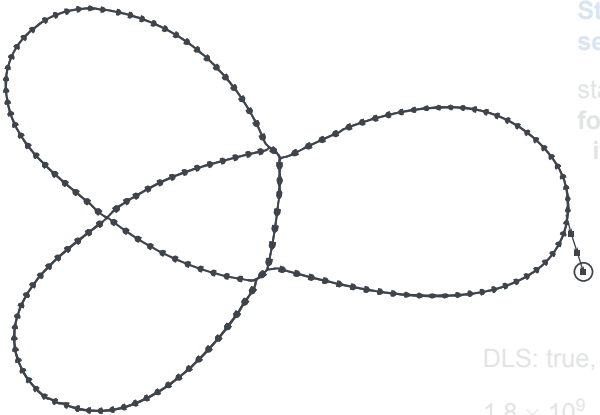
  else: stop

Cycle enumeration

CRISPR Array

DR S1 DR S2 DR S3

ATCGGCCCCGCACGCGCGGGGATGGCTC
GGGCGGCACGGCGTGTTCGCGGGGGTGGTGGCG
ATCGGCCCCGCACGCGCGGGGATGGCTC
GACGATGGCGGGGCTGTTTCCCGAGCCCTGCCGG
ATCGGCCCCGCACGCGCGGGGATGGCTC
CTCTTCGCGGGCGCCGCTGGCCGTGTCGACCAT



Start node selection

starts $\leftarrow \varnothing$
for each node v:
  if $\deg^-(v) \geq 2$:
    if DLS(v, L_{max}):
      add v to starts

DLS: true, if start = end

$1.8 \times 10^9 \longrightarrow 2.4 \times 10^4$

CRISPR array detection

Tip removal

tips : $\deg^+(v) = 0$

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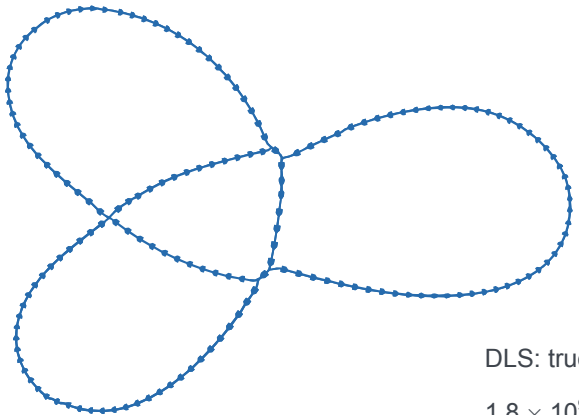
Cycle enumeration



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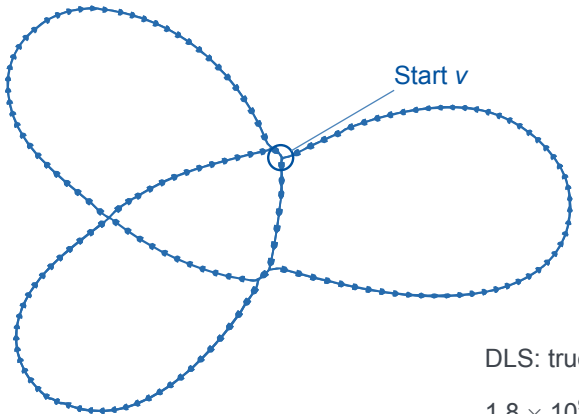

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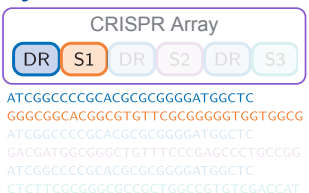

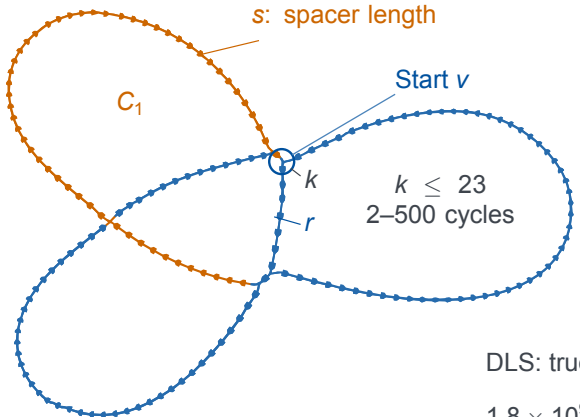

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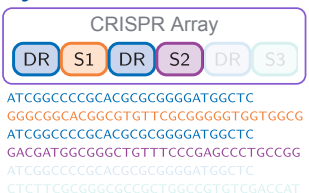

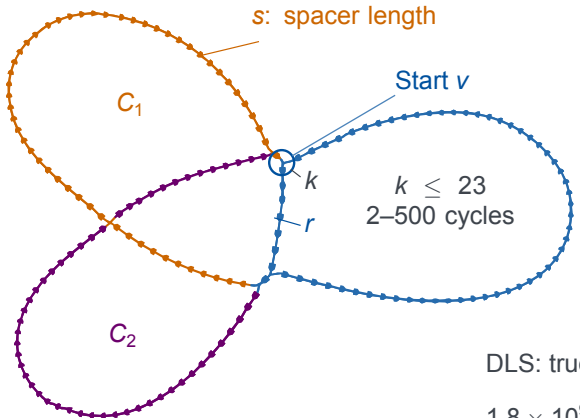

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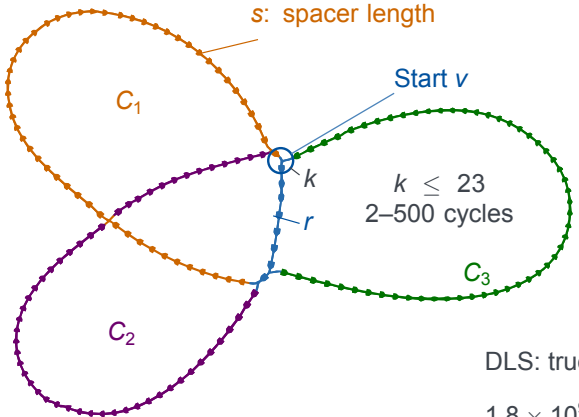

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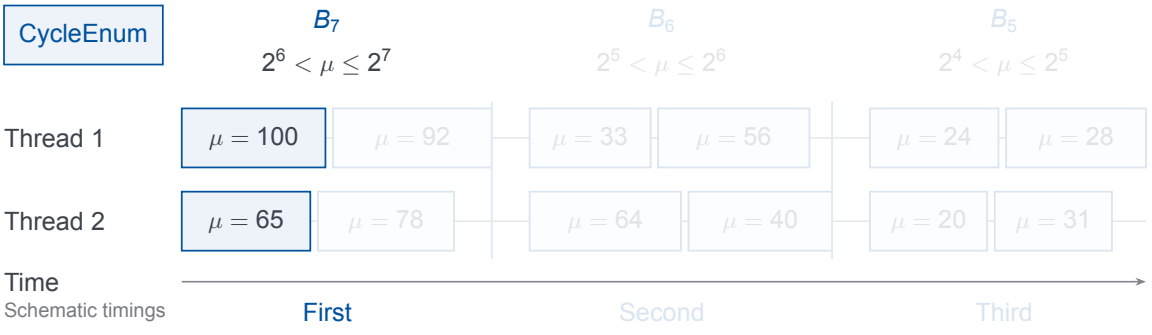
Parallelization techniques

$\mu(v)$: count of the node's k-mer

$b = \lceil \log_2 \mu(v) \rceil$

CycleEnum



High $\mu \rightarrow$ likely repeat $\rightarrow$ shared by all cycles $\rightarrow$ fewer missed cycles

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CycleEnum

B_7

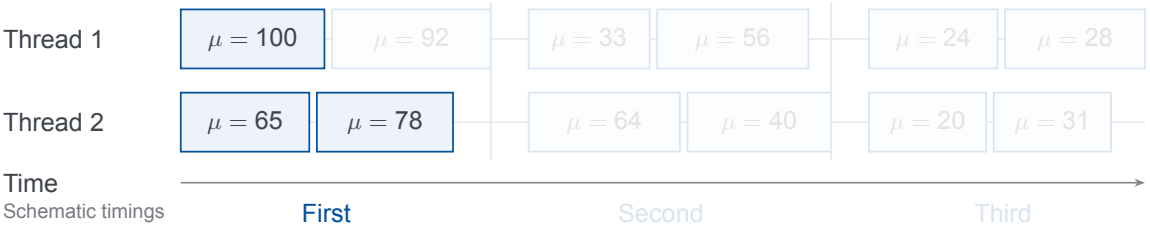
$2^6 < \mu \leq 2^7$

B_6

$2^5 < \mu \leq 2^6$

B_5

$2^4 < \mu \leq 2^5$



Schematic timings

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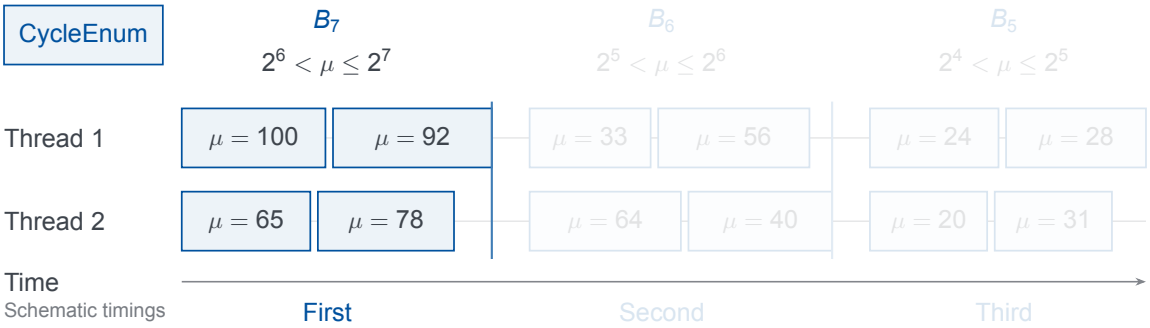

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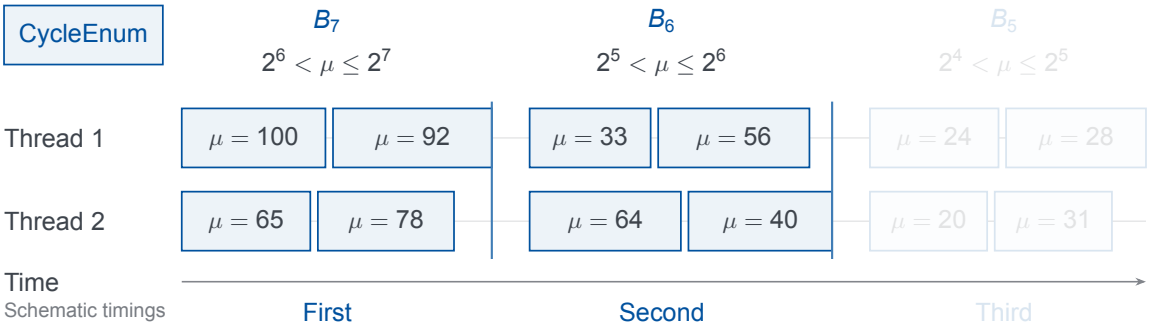

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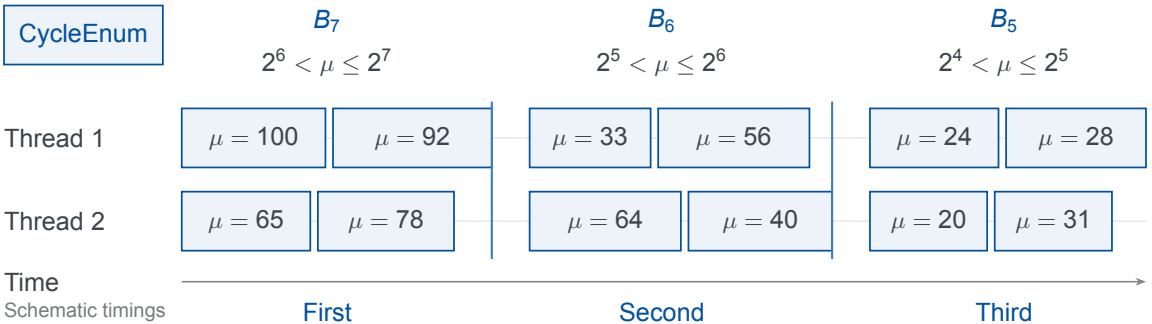

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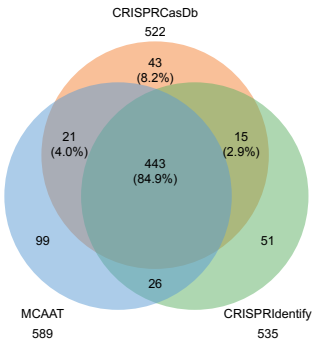
Proof-of-concept

Genome validation

57 CRISPRCasDB genomes
1–9 Mbp; ≤101 arrays/genome



Percentages of 522 reference arrays

Simulated metagenomes

P: precision; R: recall

	Arrays		Spacers	
Tool	P	R	P	R
MCAAT	0.78	0.93	0.72	0.75
CRASS	0.48	0.45	0.36	0.39
CRISPRidentify	0.57	0.67	0.43	0.32
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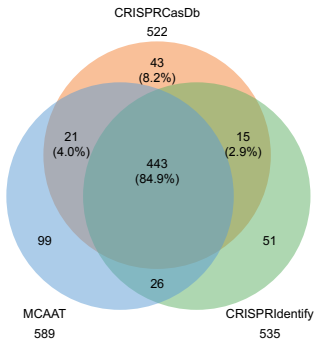

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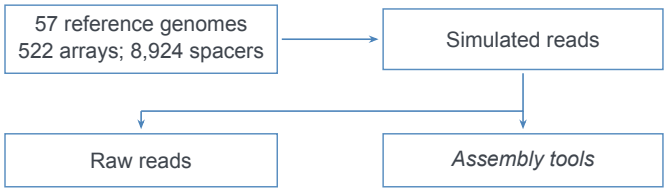

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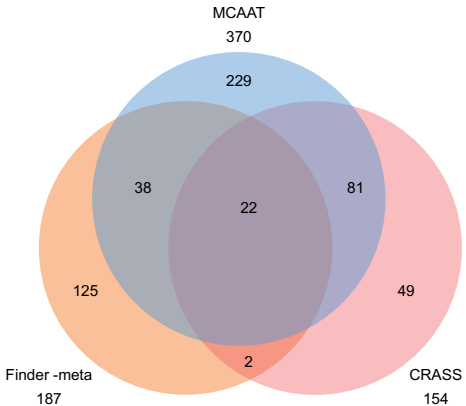
Real-world dataset

100 million metagenomic reads



- MCAAT predicts the most arrays: 370, compared with 187 and 154.
- MCAAT recovers 67% of CRASS predictions and 32% of Finder-meta predictions.
- Stronger agreement between the raw-read tools suggests an effect of the input representation.

Tool-specific predictions may extend the findings, but require independent validation.

Finder-meta: CRISPRCasFinder-meta.

Contents

- Motivation and overview
- Results I: CRISPR array detection
- **Results II: CAS cassette classification**
- Results III: Protospacer extension
- Conclusions and outlook

MCAAT 2.0: CAS cassette detection

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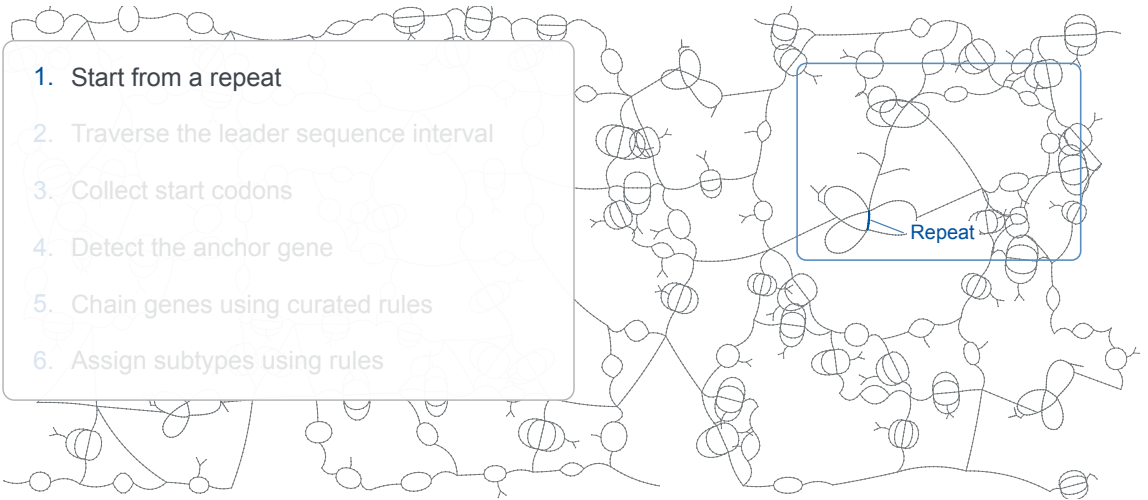
2. Traverse the leader sequence interval
3. Collect start codons
4. Detect the anchor gene
5. Chain genes using curated rules
6. Assign subtypes using rules



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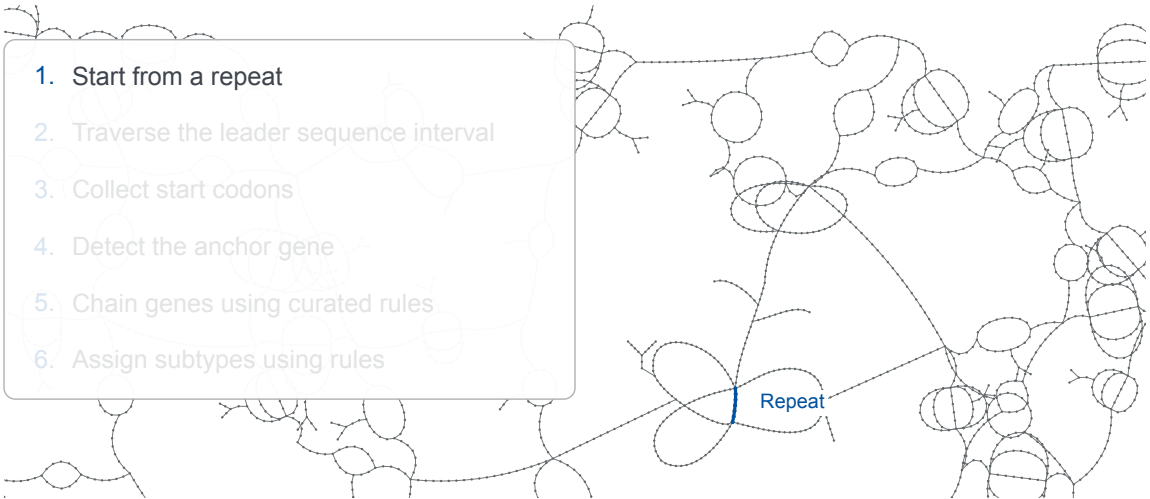

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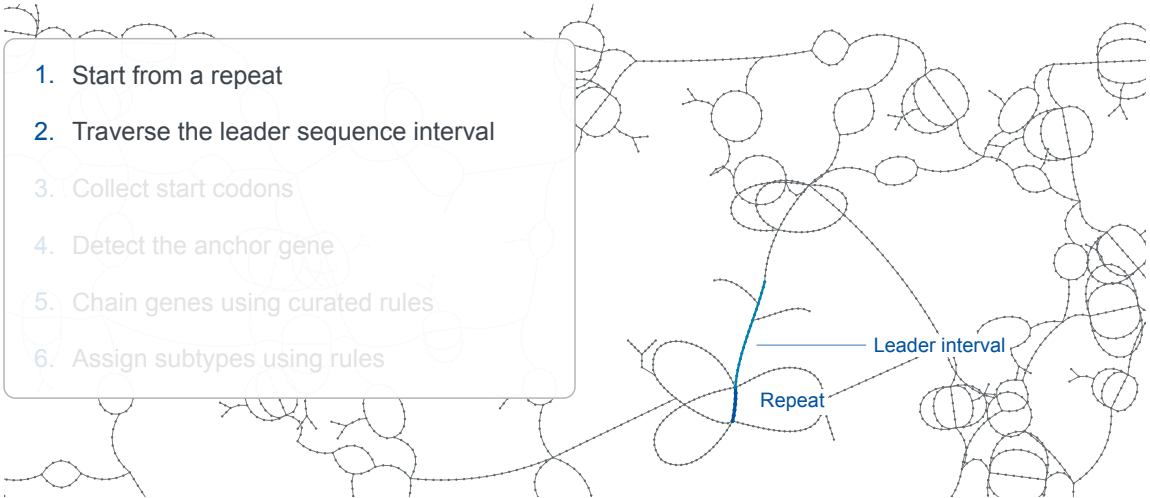

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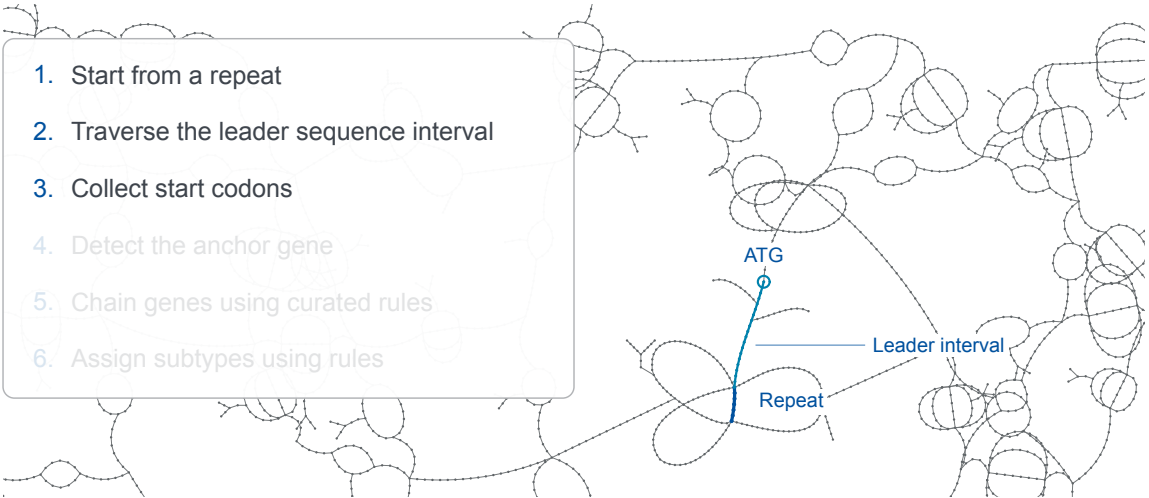

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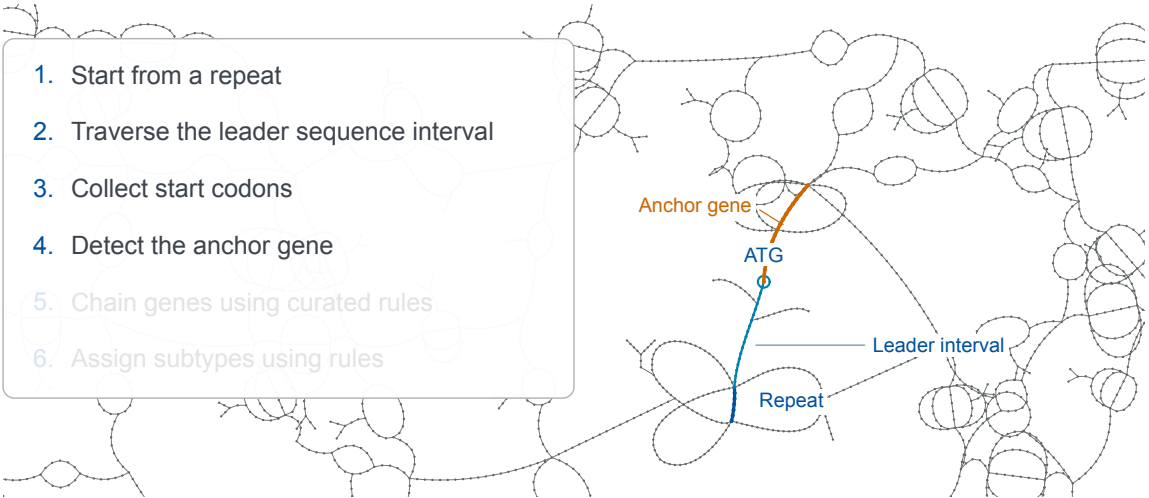

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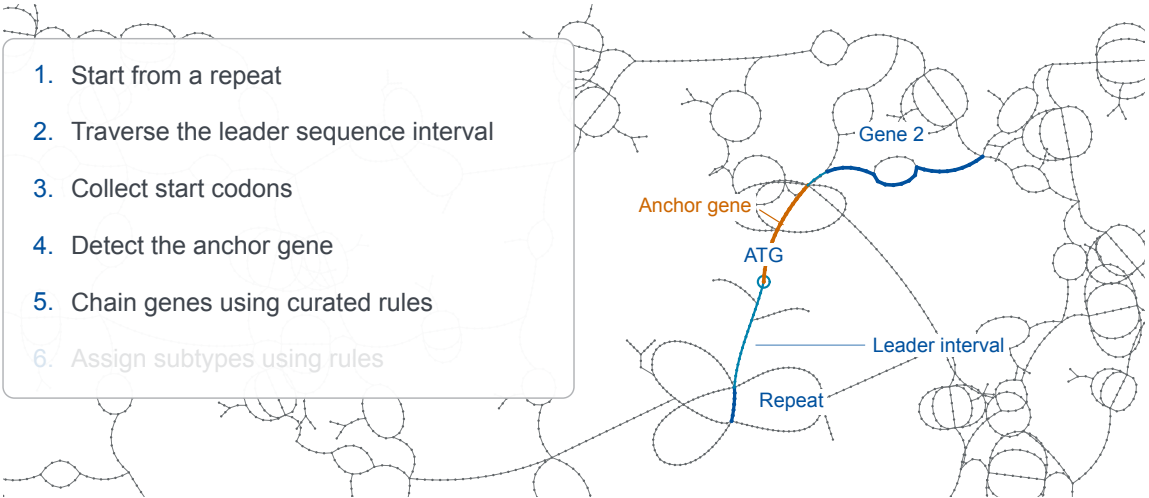

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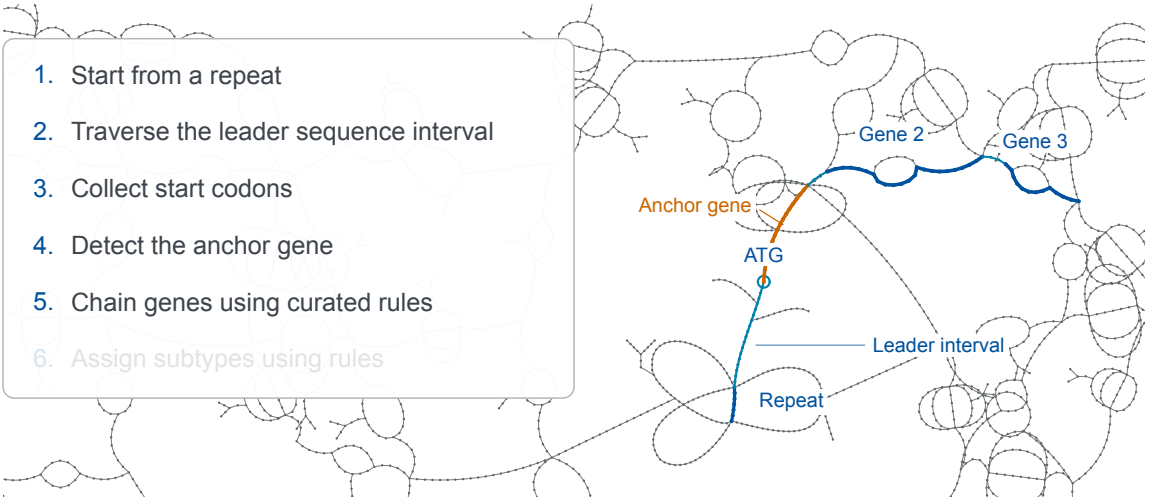

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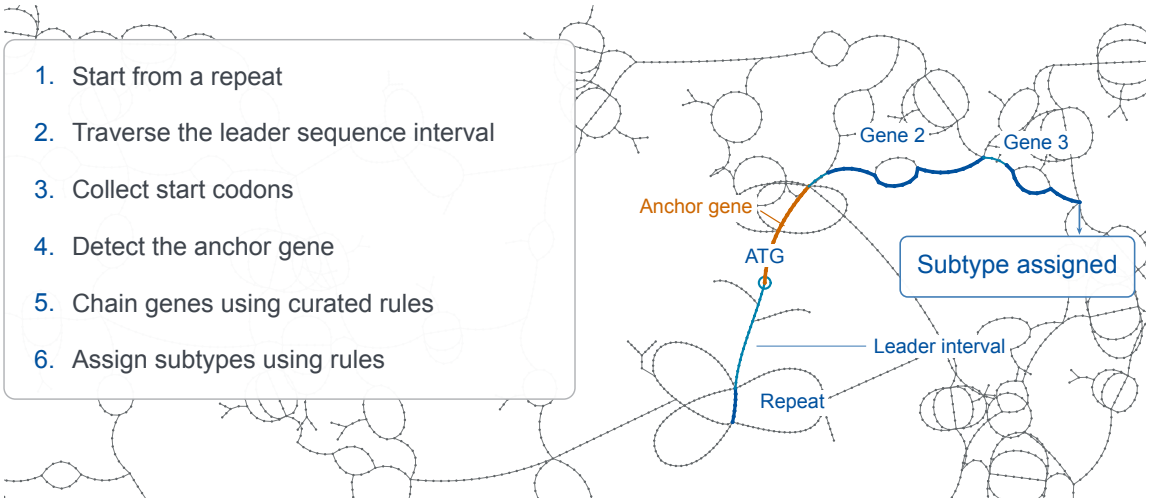

Start codon detection and profile-guided search

Start codon

ATG

Stop codons

TAA
TAG
TGA

Example codon

GCT → A

Alanine

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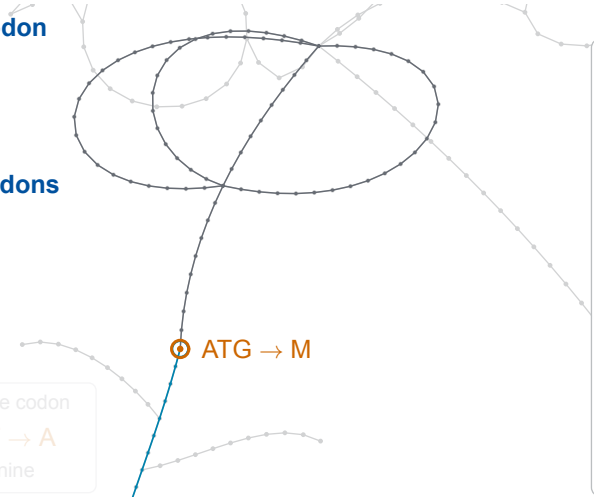

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HMM profile

Illustrative *cas2* probabilities

		Emission	Transition		
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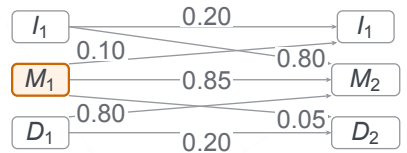
1	M	0.90	0.85	0.10	0.05
2	A	0.70	0.80	0.15	0.05
3	G	0.80	0.90	0.05	0.05
4	K	0.75	0.85	0.10	0.05
⋮	⋮	⋮	⋮	⋮	⋮

M: match I: insert D: delete



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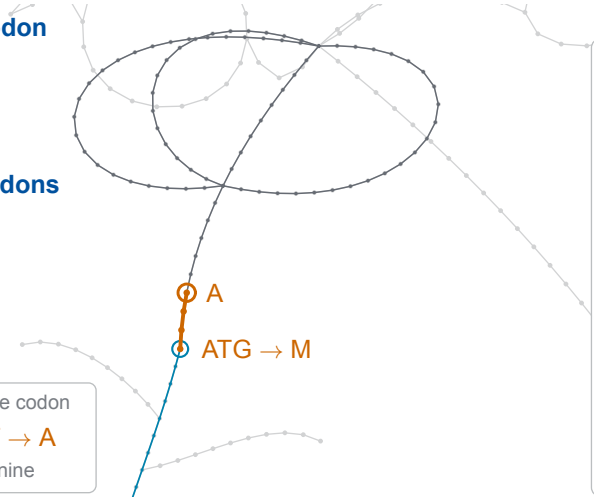

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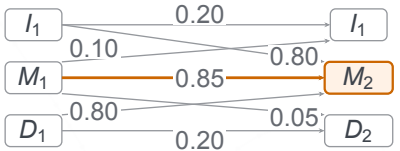

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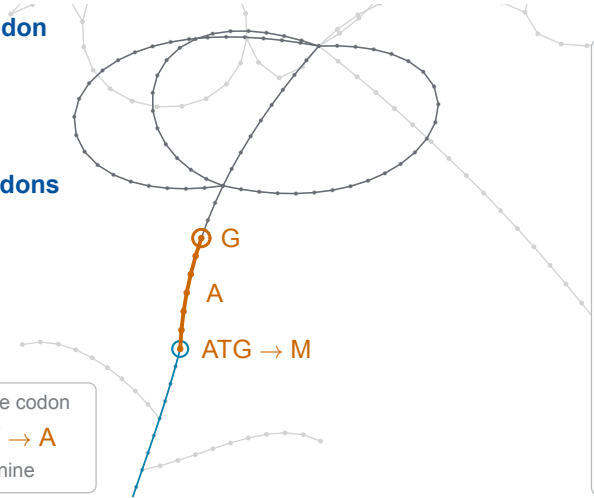

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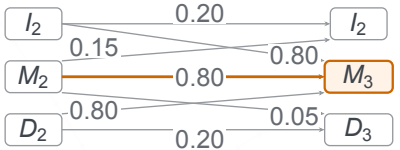

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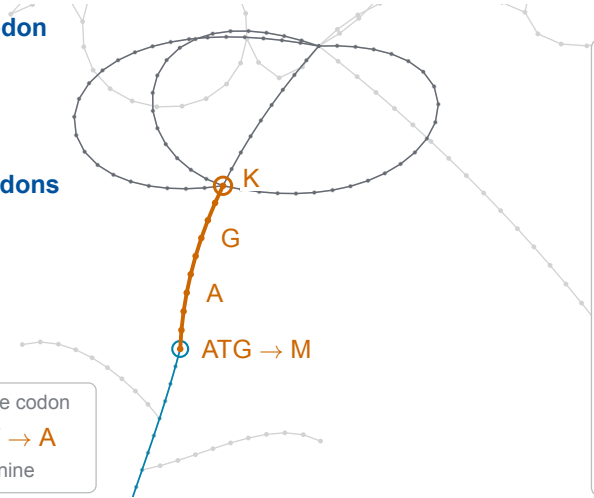

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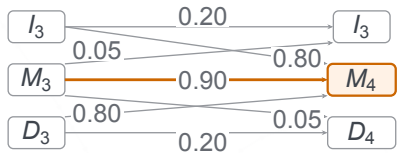

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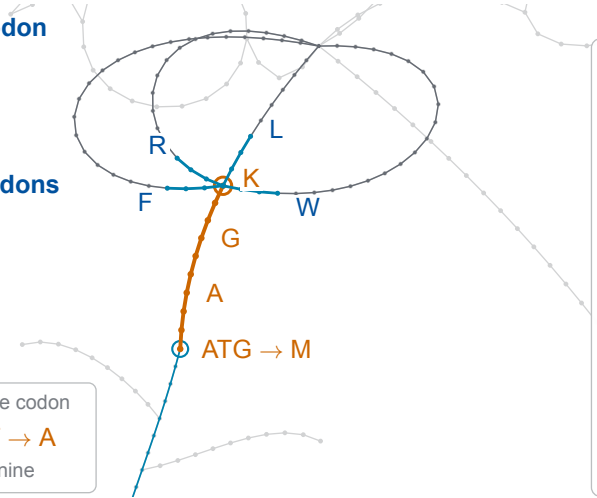

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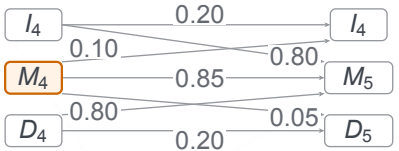


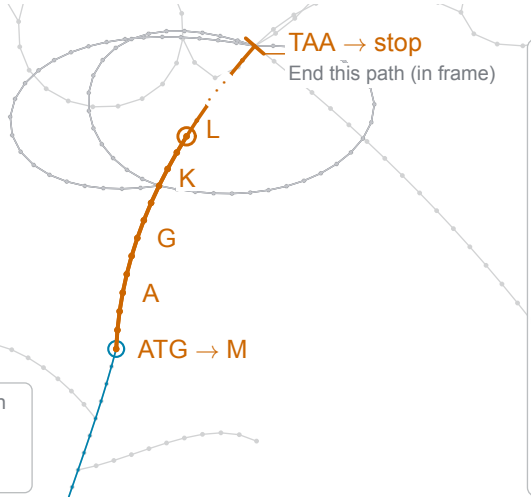
Start codon detection and profile-guided search

Start codon

ATG

Stop codons

TAA
TAG
TGA



Example codon

GCT → A

Alanine

HMM profile

Illustrative *cas2* probabilities

		Emission	Transition		
i	*a*	*P*(*a*)	*M*	*I*	*D*
1	M	0.90	0.85	0.10	0.05
2	A	0.70	0.80	0.15	0.05
3	G	0.80	0.90	0.05	0.05
4	K	0.75	0.85	0.10	0.05
⋮	⋮	⋮	⋮	⋮	⋮

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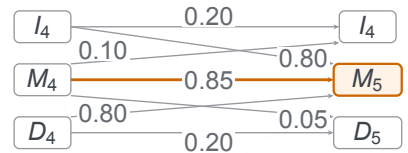

Gene chaining and subtype assignment

525 protein-sequence models (HMM profiles)

Broad family-specific: **80**

37 subtypes

1. Search from ATG

Two possible next genes

Family	I-E compatible
cas8e	
cas5	
cas7	

Gene chaining and subtype assignment

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2. Find a *cas8e* candidate

Two possible next genes

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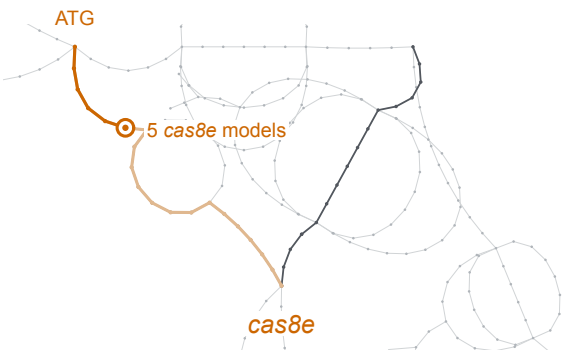
3. Refine at the same ATG



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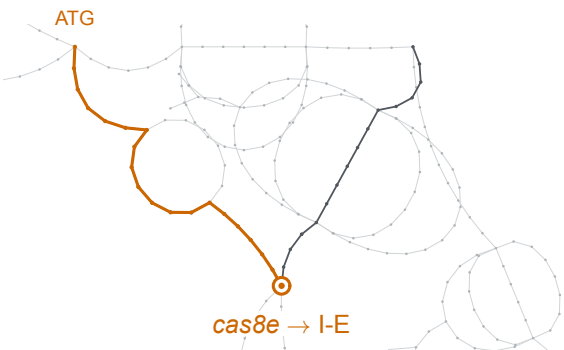
4. Apply subtype I-E rules



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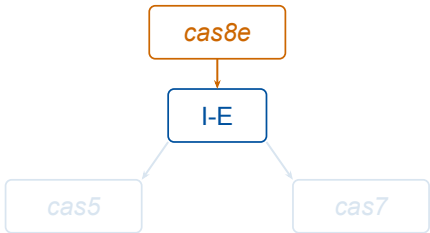

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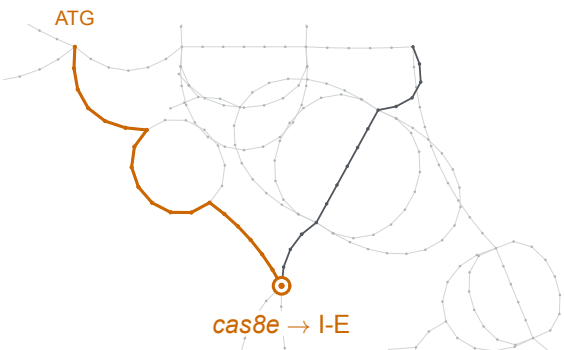

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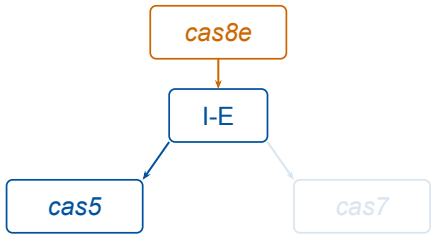

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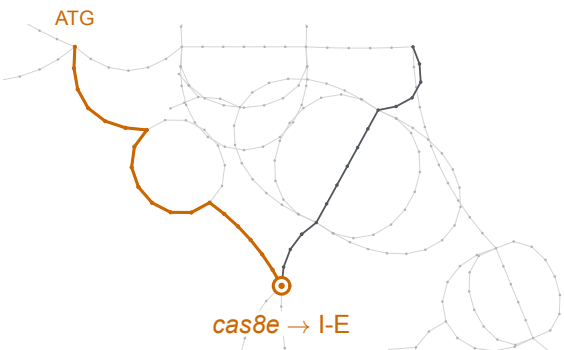

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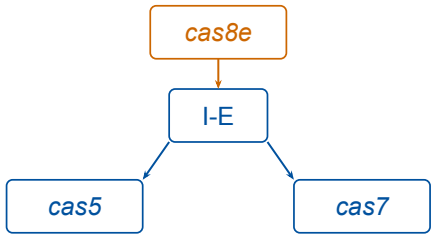

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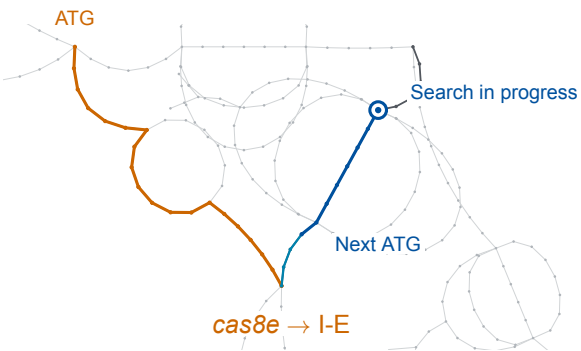
5. Search the next gene



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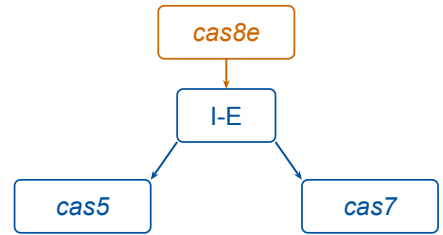

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Proof-of-concept in a simulated metagenome

- Simulated metagenome from 24 CRISPRCasDB genomes.
- 525 CASFinder profiles and subtype rules.
- Reference: MacSyFinder on each genome separately.

Reference systems	75
MCAAT predictions	114
Matched systems	62

Recall: 82.7% Precision: 54.4%
F1: 65.6%

Additional predictions require validation.

The contig-based comparison was stopped after seven days.

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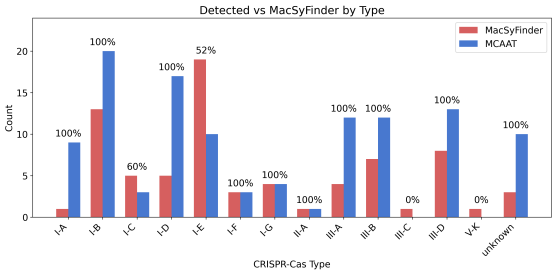

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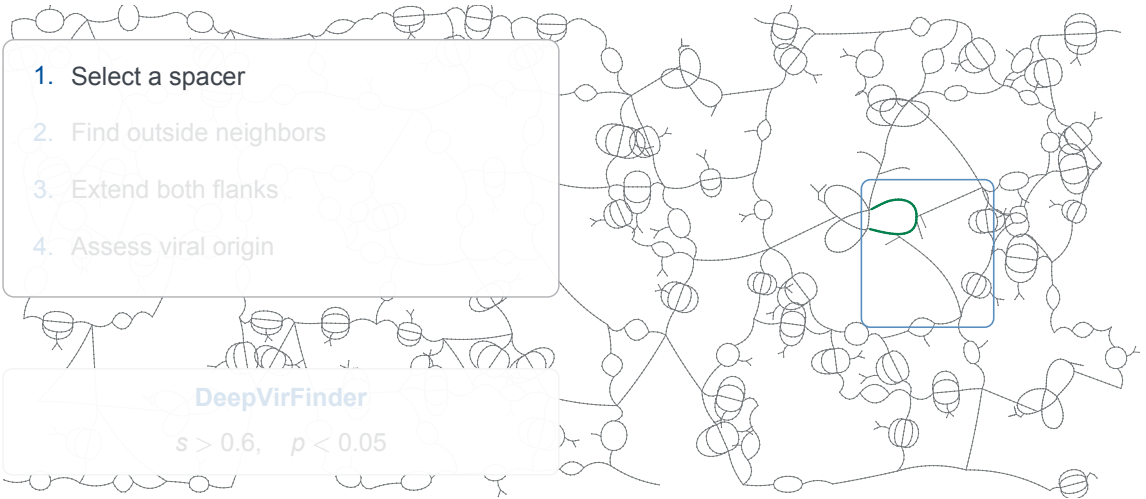
- **Results III: Protospacer extension**
- Conclusions and outlook

Protospacer detection and extension

1. Select a spacer
2. Find outside neighbors
3. Extend both flanks
4. Assess viral origin

DeepVirFinder

$s > 0.6, \quad p < 0.05$



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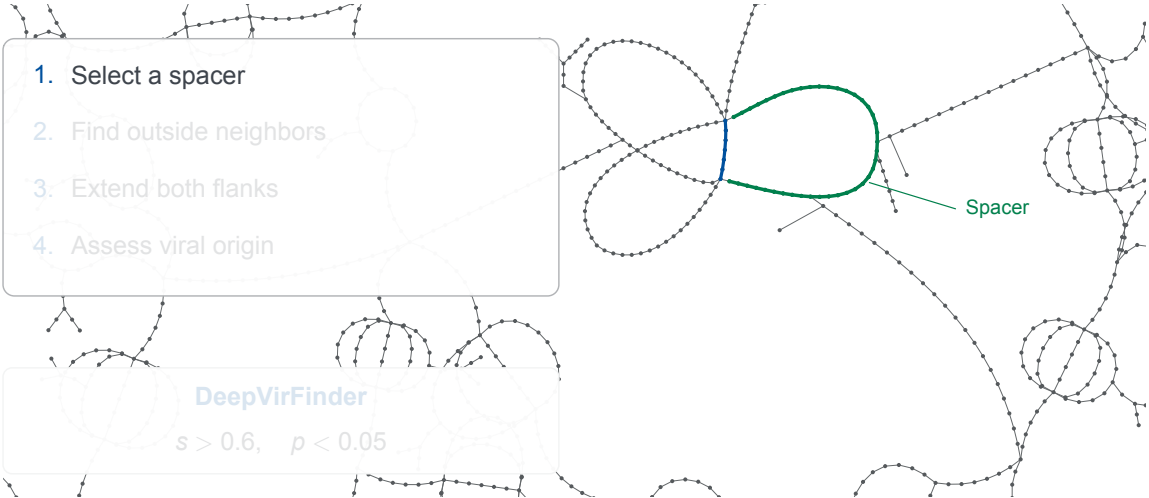

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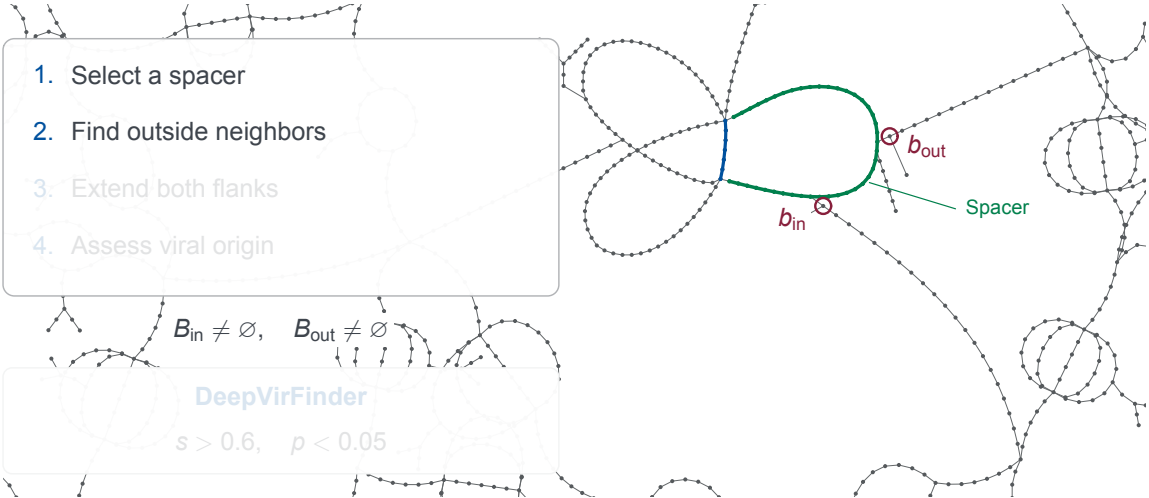

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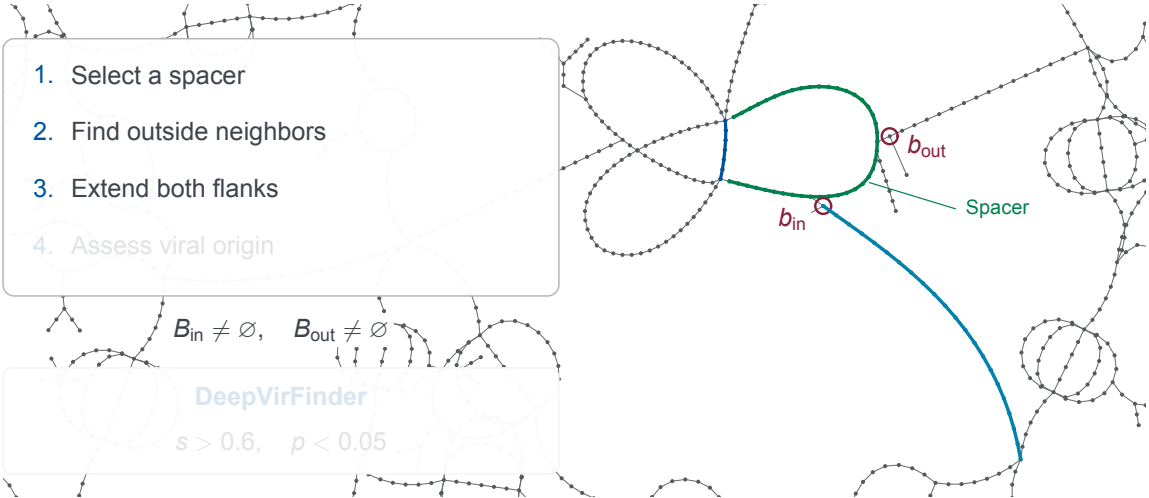

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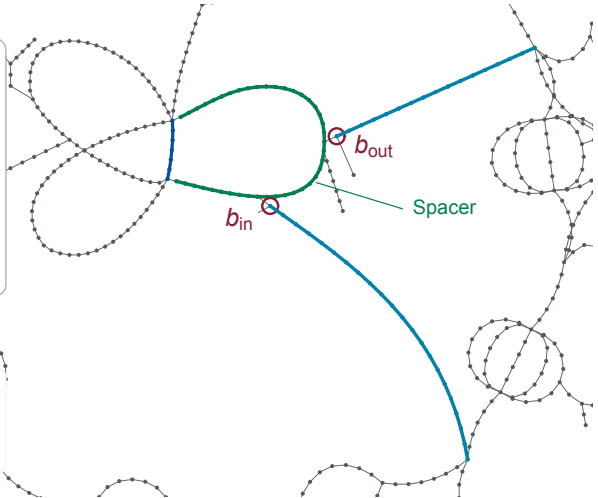

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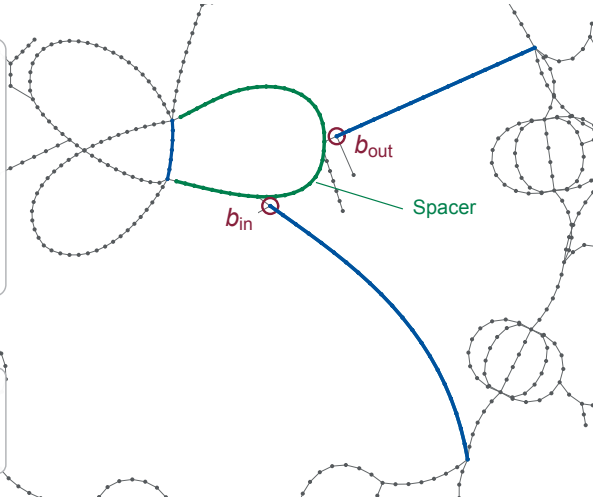
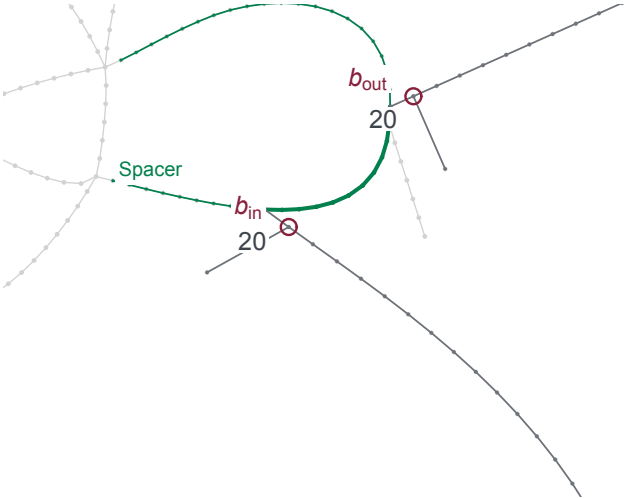


Coverage-guided extension



Extend outside the tangle

$0.1\mu(b) \leq \mu(v) \leq 5\mu(b)$

$\mu(b) = 20 \Rightarrow [2, 100]$

Mean multiplicity

$$s_n = \frac{(n-1)s_{n-1} + \mu(v_n)}{n}$$

Keep 50 best paths

Up to 1,250 bp per flank

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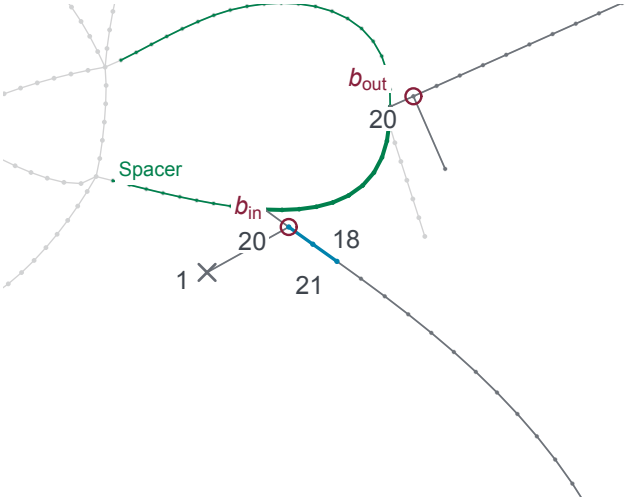

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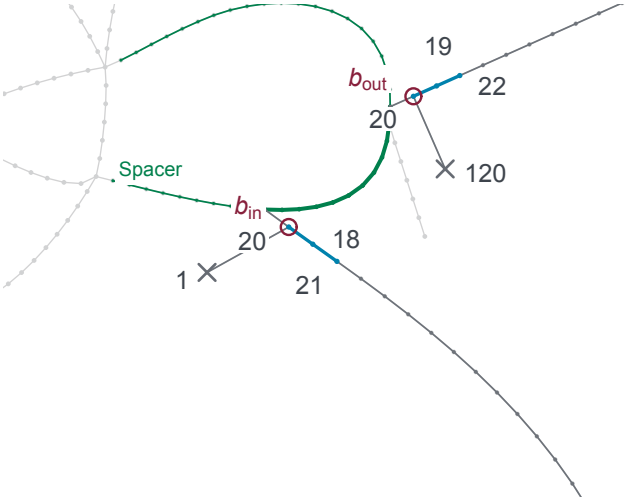

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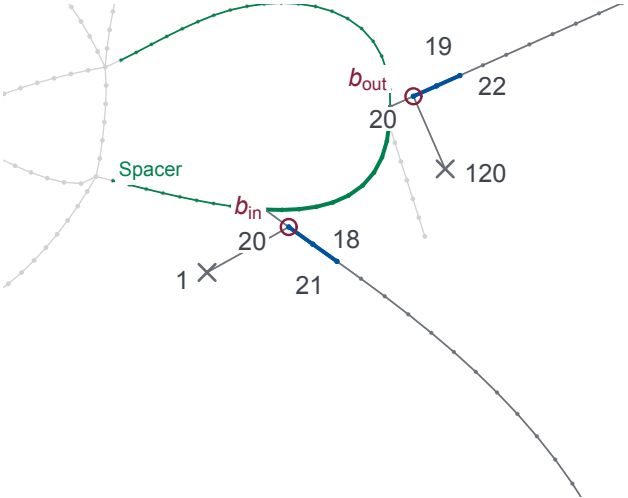

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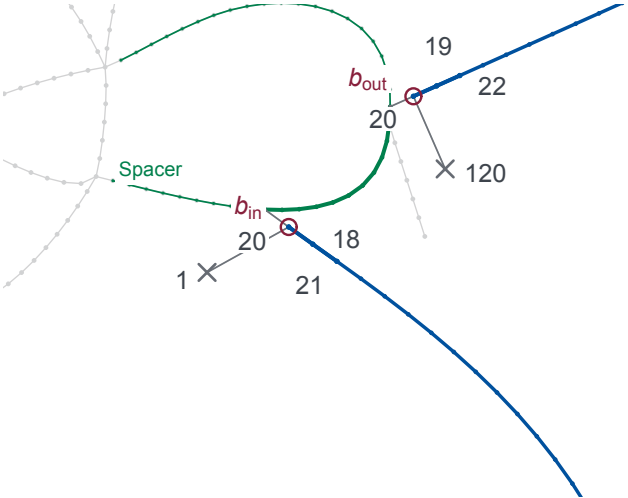

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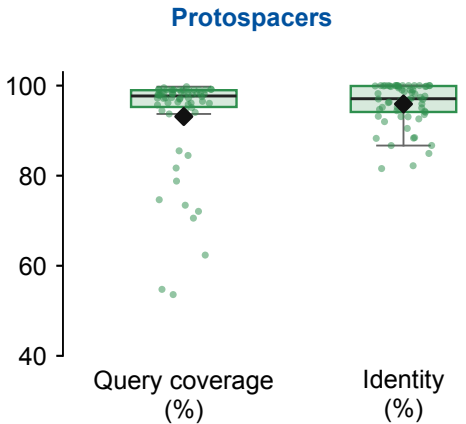
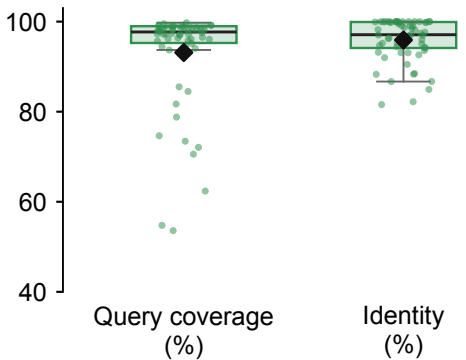


Data and comparison

- **3** human gut metagenomes: Shah et al. (2023), PRJEB46943.
- MCAAT: **264** CRISPR arrays with **2,747** spacers.
- **236** protospacers: BLAST against Shah's phage catalogue, then NCBI nt.
- Viral database matches: **85/236 (36.0%)**.

Proof-of-concept

Protospacers



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Conclusions and future outlook

Raw reads → de Bruijn graph

CRISPR arrays

93% recall
78% precision

Simulated metagenome

CAS cassettes

62/75 systems recovered
114 predictions

Simulated metagenome

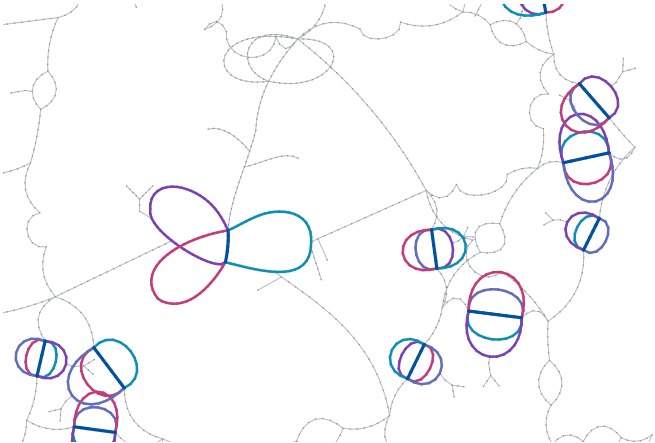
Protospacers / phages

85/236 protospacers
64/236 extensions

Viral DB matches (real metagenomes)

Conclusions and future outlook

Cycle enumeration and array detection



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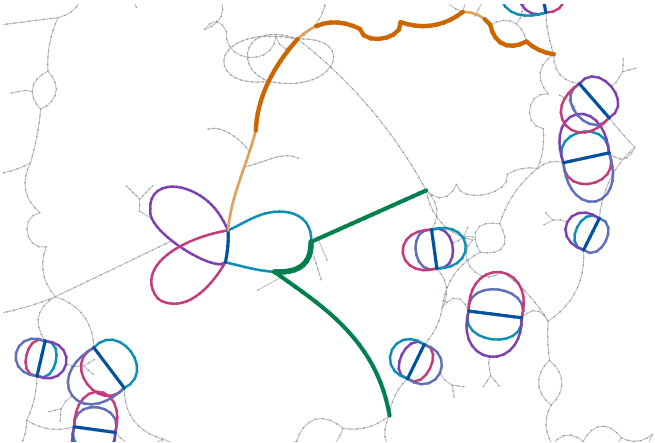

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Future outlook

- Reduction of false positives in CRISPR array detection
- Reconstruction of CRISPR array and spacer ordering
- Study of spacer evolution using SpacerPlacer on ordered CRISPR arrays
- Large-scale analysis of metagenomes for CRISPR arrays, CAS genes and protospacers
- Building metagenome database similar to CRISPRCasDB

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Thank you!

Prof. Dr. Björn Voß, Prof. Dr. Franz Baumdicker,
Prof. Dr. Stefan Legewie,
Prof. Dr. habil. Martin Siemann-Herzberg,
Jun.-Prof. Dr. Nieves Peltzer, Prof. Dr. Christina Wege,
Prof. Dr.-Ing. Ralf Takors

Mama, Papa, Farid, Elten Polukhov, Richard Schäfer,
Pavan Kumar, Christopher Sarkizi, Sidharth Jayasankar,
Andreas Ulmer, Charlene Wüllenrath, Caroline Sylvester,
Christopher Adelmann, Marius Sebeke, Qian Xiong,
Tobias Müller, Jan Spindler, Maren Lösch, Katharina Hofer,
Christoph, Arnav Upadhyay, Steven Minden, Atilla Teleki,
Gennaro Avolio, Guillermo Tapia, Carlos Castillo, Russel Ngo

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German Research Foundation

SPP 2141

CRISPR-Cas functions
beyond defence